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應用科學學院
Faculdade de Ciências Aplicadas
Faculty of Applied Sciences

Doctor of Philosophy in Computer Applied Technology

Thesis Defence Examination

Thesis Review Comments

Student

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Title of Thesis	Improving the Reliability and Accuracy of Dialogue-Based Natural Language Generation		
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Reviewer

Name	Prof. Bo Liu	Date	2026-05-26
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Recommendation

☒ Accepted ☐ Suggestion for Revision*

* "Suggestion for Revision" will result in the delay of the oral defence.

Comments

Overview.

This thesis presents a comprehensive study on improving the reliability and accuracy of dialogue-based natural language generation, with dialogue summarization as the primary experimental focus. The work is timely, technically relevant, and well aligned with current research trends in pretrained language models, dialogue understanding, and trustworthy generation. The thesis addresses several important challenges in dialogue-based generation, including role-sensitive modeling, interleaved topic structure, dynamic semantic evolution, hallucination mitigation, and the practical limitations of reasoning-oriented large language models.

The overall structure is logical and progressively organized. The research evolves from role-aware dialogue modeling (RAA), to hierarchical topic disentanglement (HiSum), then to dynamic semantic adaptation (DABART), and finally to reliability-oriented generation and reasoning-model evaluation (SSCM and reasoning LLM analysis). This progression demonstrates a coherent long-term research direction rather than a collection of unrelated

studies.

The thesis is generally well written, technically mature, and experimentally extensive. The inclusion of both English and Chinese datasets, multiple evaluation protocols, human evaluation, ablation studies, and qualitative analysis strengthens the empirical credibility of the work. The thesis also demonstrates awareness of practical deployment concerns in high-stakes domains such as finance and healthcare.

A minor weakness is that some chapters remain strongly engineering-oriented, with incremental transitions between methods. While the thesis consistently improves dialogue summarization performance, the broader unifying theoretical framework connecting all proposed methods could be articulated more explicitly in later chapters and in the conclusion.

Knowledge Base and Literature Review.

The thesis demonstrates a solid and up-to-date understanding of the dialogue summarization and dialogue-grounded generation literature. The literature review is well organized into generative paradigms, structural modeling, and reliability/faithfulness analysis.

The discussion of pretrained architectures, PEFT methods, prompting strategies, and reasoning-oriented LLMs reflects good coverage of contemporary NLP developments. The review successfully contextualizes the research within the transition from traditional extractive methods to Transformer-based and LLM-based paradigms.

Particularly commendable is the inclusion of reliability-oriented generation and hallucination mitigation as a core literature component rather than a peripheral discussion. The thesis correctly identifies that fluent generation does not necessarily imply faithful generation, which is an important contemporary issue in large language models.

However, several aspects could be improved: The review focuses predominantly on dialogue summarization literature and less on broader discourse modeling, conversational pragmatics, or discourse planning theories that could strengthen the conceptual foundations. The reliability discussion would benefit from deeper engagement with factuality evaluation literature beyond summarization-specific hallucination studies. More discussion of recent retrieval-augmented generation and verifier-based generation frameworks could strengthen the positioning of SSCM relative to broader trustworthy generation research.

Overall, the literature review is comprehensive, current, and sufficient for a PhD thesis in this field.

Originality and Independence.

The strongest aspect of originality lies not necessarily in any single architectural novelty, but in the sustained and coherent focus on dialogue-specific reliability and structural modeling problems. The work consistently emphasizes that dialogue differs fundamentally from monologic text due to distributed semantics, speaker interaction, and evolving contextual

salience. This conceptual framing gives the thesis a coherent identity.

The SSCM chapter is particularly noteworthy because it extends beyond standard summarization optimization and addresses reliability-sensitive generation under weak supervision. The self-supervised ranking strategy for hallucination mitigation is practically meaningful and shows independent research thinking.

The reasoning-oriented LLM evaluation chapter also contributes originality through empirical analysis rather than method design. The finding that reasoning-oriented models do not consistently outperform non-reasoning models in dialogue summarization is interesting and potentially impactful.

The thesis demonstrates good independence overall. The work spans methodological design, benchmarking, evaluation methodology, and empirical analysis. The candidate appears capable of identifying research problems, designing solutions, and critically evaluating broader trends in the field.

One limitation is that some proposed methods resemble natural extensions of prior graph-based or hierarchical dialogue summarization frameworks rather than fundamentally new paradigms. The originality is therefore stronger at the system-integration and problem-formulation level than at the level of entirely novel modeling theory.

Arguments and Reasoning.

The thesis also demonstrates good scientific discipline in validating individual module contributions through ablation experiments. The inclusion of computational analysis and convergence analysis is appreciated because it reflects awareness of practical deployment considerations.

The reasoning in the final evaluation chapter is especially strong. Rather than assuming reasoning-based models are universally superior, the thesis empirically investigates their behavior and identifies cases where explicit reasoning may introduce redundancy or structural inefficiency. This critical stance is valuable and scientifically mature.

Nevertheless, several weaknesses remain: Some claims regarding “reliability” are still primarily supported through summarization metrics and human judgments rather than formal factuality verification frameworks. The conceptual distinction between “semantic salience,” “structural coherence,” and “faithfulness” could be more rigorously defined. Some methodological transitions appear incremental, and stronger theoretical synthesis across chapters would improve the argumentative depth.

Despite these limitations, the thesis demonstrates strong analytical capability and sound experimental reasoning overall.

Evaluation within the Field.

The empirical evaluation is extensive. The thesis includes multiple benchmark datasets across Chinese and English domains, including customer service, meetings, medical dialogue, and financial tasks.

The work also compares traditional seq2seq models with contemporary LLMs such as GPT-4 and reasoning-oriented systems, which increases the relevance of the evaluation to the current state of the field.

From a publication-quality perspective, the work appears to meet the standard expected of a PhD thesis in NLP or dialogue systems research. The research contributions are sufficiently substantial and experimentally validated.

The most impactful component may ultimately be the evaluation and analysis of reasoning-oriented LLMs, because this topic is highly contemporary and the conclusions challenge prevailing assumptions in current LLM discourse.

One concern is that several chapters optimize relatively similar summarization settings and metrics. The thesis could potentially benefit from stronger external validation through downstream user studies, deployment-oriented evaluation, or broader generalization experiments beyond summarization-centric benchmarks.

Completeness and Coherence.

The organization is coherent and easy to follow. The progression from dialogue structure modeling toward reliability and reasoning analysis creates a strong narrative continuity. The repeated focus on dialogue-specific challenges helps unify the thesis.

The experimental sections are thorough, with adequate baselines, human evaluations, ablations, visualizations, and case studies.

The writing quality is generally high. The thesis is clearly written, technically precise, and professionally formatted.

Minor issues include: Some methodological overlap across Chapters 4–6 creates partial redundancy. The conclusion chapter could provide a stronger higher-level synthesis of the proposed frameworks instead of mainly summarizing chapter outcomes. The broader implications for trustworthy LLM design beyond dialogue summarization could be expanded further.

Overall, the thesis is complete, coherent, and mature.

Conclusion.

The thesis makes meaningful contributions in dialogue summarization, structural dialogue modeling, hallucination mitigation, and reasoning-oriented LLM evaluation. The

experimental evaluation is extensive and generally convincing, and the candidate demonstrates the ability to conduct independent research at a high level.

Although some contributions are more incremental than fundamentally paradigm-shifting, the overall body of work is substantial and well executed. The thesis meets the standard expected for a doctoral dissertation in natural language processing and dialogue systems research.

The thesis is recommended for approval, subject to normal oral defense examination and possible minor revisions regarding theoretical synthesis and positioning relative to broader trustworthy generation literature.

Additional Comments.